

# Hands-On-Learning: Agent Typology

## Explorer: Classify and Map Agent Roles

# Description

Learners will identify, classify, and visually map different agent types (reactive, deliberative, hybrid) and their typical use cases in a real-world scenario. They will create a simple interaction map showing how these agents cooperate or compete in a specific domain, and submit both their classification and reasoning for AI grading.

# Learning Objectives

By the end of this activity, learners will be able to:

1. **Classify** agents into key categories (reactive, deliberative, hybrid).
2. **Describe** typical tasks, strengths, and weaknesses of each agent type.
3. **Construct** a simple interaction map illustrating agent relationships in a chosen scenario.
4. **Analyze** how agent type selection affects system performance, complexity, and adaptability.



# Assignment Components

## Introductory Slide (Visually Engaging)

- **Title:** Agent Ecosystem Mapping — Who Does What?
- **Subtitle:** "Visualizing agent types and their roles in a shared environment."

## Tools to be used:

- **Diagram Tool:** draw.io
- **Text Editor:** Google Docs / Notion / Any Markdown editor for short write-ups.

# Scenario

You are a systems analyst at an urban traffic control center preparing to integrate AI agents into smart intersections. The city wants to optimize traffic flow while reducing accidents and emissions. Your job is to classify three proposed agents - Sensor Agent (real-time data collector), Traffic Light Controller (actuator), and Route Planner (navigator) - into correct agent types, map their interactions, and explain why this classification matters for city performance.



# Instructions for Learners (Step-by-Step)

1

- Review the Agent Types  
Skim the reference table of AI agent types (reactive, deliberative, hybrid).  
**Outcome:** Ready to classify based on key attributes.

2

- Classify the Agents  
Decide which type each proposed agent belongs to (justify in one sentence each).  
**Outcome:** Table with agent name, type, and short justification.

3

- Map the Interactions

Use a diagram tool to create a simple flow: how data moves from Sensor Agent → Traffic Light Controller → Route Planner and back (if feedback loop exists).

Annotate arrows with short labels (e.g., “traffic data,” “control command”).

**Outcome:** Interaction diagram.

3

- Analyze Implications

Write 100–150 words on: How does using reactive vs deliberative agents affect speed, accuracy, and system resilience?

**Outcome:** Short write-up highlighting trade-offs.

# Deliverables for Submission (AI Grading)

## Deliverable 1: Primary Work / Creation

→ **Question Prompt:** "Upload your classified agent table and interaction diagram showing agent roles and data flow."

→ **Response Type:** File Upload

→ **File Types:** PNG, PDF, PPTX, DOCX



## Deliverable 2: Process / Methodology

- **Question Prompt:** "Briefly describe how you decided on each classification (mention key attributes you used), your reasoning"
- **Response Type:** File Upload or Rich Text
- **File Types:** TXT, PDF, DOCX

## Deliverable 3: Analysis / Critical Thinking Component

→ **Question Prompt:** "Explain in 100–150 words how the choice of agent types affects speed, scalability, and decision quality in your traffic system."

→ **Response Type:** Rich Text

→ **File Types:** TXT, PDF

## Deliverable 4: Reflection / Learning Insights

- **Question Prompt:** "In a few sentences describe, what surprised you about classifying and mapping these agents, and what would you do differently in a real deployment?"
- **Response Type:** File Upload
- **File Types:** TXT, PDF, MP3, MP4

## Best Practices for AI-Graded Submission

- **Specificity:** Use agent type names explicitly (Reactive, Deliberative, Hybrid).
- **Clarity:** Diagrams must have labeled nodes and arrows — AI grading looks for connectivity.
- **Depth:** Go beyond surface-level by discussing trade-offs (e.g., "reactive agents are fast but brittle").
- **Progression:** Submit diagram first, then reasoning and reflection.

# Mapping Core Capabilities of a Personal AI Task Agent



Learners will explore the foundational concepts of AI agents by modeling the capabilities of a simple personal task agent, such as one that manages daily schedules, to understand how perception, reasoning, and action enable autonomous behavior. This hands-on activity involves diagramming agent components and documenting their interactions in a simulated scenario, providing a practical introduction to agent design without extending to multi-agent coordination or governance.

# Learning Objectives

1. Identify and define at least three core capabilities (e.g., perception, reasoning, action) of a single AI agent.
2. Diagram how these capabilities interact to achieve a specific task goal in a personal assistant context.
3. Explain the role of autonomy and reactivity in enabling an agent's environmental adaptation.
4. Apply basic agent concepts to evaluate a simple scenario's requirements.



# Assignment Components

## Introductory Slide

**Title:** Unlocking Autonomy: Building Your Personal AI Task Manager's Core Engine • **Subtitle:** Diagram the Perception-Reasoning-Action Cycle for Everyday Efficiency

## Tools to be used:

Diagramming Software (e.g., Draw.io, Lucidchart, or Microsoft Visio for creating flowcharts)

Text Editor (e.g., Google Docs, Microsoft Word, or Notion for documentation)



# Scenario

You're a busy freelance writer in a vibrant creative hub like Austin, Texas, juggling deadlines, meetings, and personal errands amid a hectic schedule. Your current calendar app alerts you to conflicts but can't proactively suggest adjustments or integrate weather updates for outdoor tasks. The challenge is to conceptualize a single AI agent that autonomously monitors your day, interprets changes (e.g., a sudden rain delay), and takes initiative to reschedule, all while operating independently without human oversight.

# Instructions for learners

## 1 Define Core Capabilities

→ List and define three core AI agent capabilities (e.g., perception for input sensing, reasoning for decision-making, action for output execution) relevant to schedule management.

→ **Expected Outcome:** A short paragraph (100-150 words) outlining each capability with a one-sentence definition and example tied to the scenario (e.g., perception detecting a weather API alert).

→ **Tip:** Draw from Module 1's reading "The Evolution of AI Agents" for precise terminology; avoid referencing multi-agent interactions.

## 2 Diagram Capability Interactions

- Create a simple flowchart showing how the three capabilities interact in a cycle to handle a scenario event, such as rescheduling a rain-affected outdoor meeting (start with perception, flow to reasoning, end with action).
- Include arrows for data flow and brief annotations (1-2 sentences per capability).
- **Expected Outcome:** A visual diagram illustrating the agent's autonomous loop for one task resolution.
- **Warning:** Keep the diagram to 4-6 elements to maintain simplicity; no need for code or advanced simulations.

### 3 Evaluate Autonomy and Reactivity

→ Write a 100-word explanation of how autonomy (goal-directed independence) and reactivity (environmental responsiveness) enhance the agent's effectiveness in the scenario, using your diagram as reference.

→ **Expected Outcome:** A focused analysis highlighting one strength and one potential limitation of these traits in a personal context.

## 4 Reflect on Application

Jot down 50-75 words on how this agent's core concepts could adapt to a different daily challenge, like email prioritization.

**Expected Outcome:** A brief insight connecting the activity to broader personal productivity.



# Deliverable for Submission (AI Grading)

## 1 Deliverable 1: Capability Definitions

→ **Question Prompt:** "Submit your 100-150 word paragraph defining three core AI agent capabilities (perception, reasoning, action) with scenario-specific examples."

→ **Response Type:** File Upload

→ **File Types:** .docx, .pdf, .txt

## 2 Deliverable 2: Interaction Flowchart

→ **Question Prompt:** "Upload your flowchart diagramming the interaction of the three capabilities in a perception-reasoning-action cycle for rescheduling a meeting."

→ **Response Type:** File Upload

→ **File Types:** .png, .jpg, .pdf

### 3 Deliverable 3: Autonomy Evaluation

→ **Question Prompt:** "Provide a 100-word analysis explaining how autonomy and reactivity improve the agent's task handling, including one strength and one limitation with diagram references."

→ **Response Type:** Rich Text

→ **File Types:** Not applicable (text input)



## 4 Deliverable 4: Application Reflection

**Question Prompt:** "Reflect in 50-75 words on adapting these agent concepts to another personal challenge, like email prioritization."

**Response Type:** File Upload

**File Types:** .docx, .pdf, .txt, .mp4, .mp3

# Share The Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

# Instructions for Documenting and Sharing The Project for Peer Review

## 1. Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

## 2. Share the Project

- Save your report in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

# Instructions for Documenting and Sharing The Project for Peer Review

## 3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.